ROBIN SAINI

Lead / Principal Data Engineer | Snowflake Data Platforms & Agentic AI | Python & SQL | PySpark | Airflow & dbt
London, UK | robinsaini156@gmail.com | +44 7466 201865 | linkedin.com/in/robinsaini | github.com/cloudcruncher
Right to work in the UK (ILR) | Eligible for SC clearance

PROFESSIONAL SUMMARY

Lead data engineer in regulated UK banking: 15+ years in software and data engineering, 8+ in data engineering. Hands-on in Python, SQL, Snowflake, PySpark, Airflow and dbt.

- **Now:** lead data engineer for NatWest's enterprise innovation team, taking agentic AI products from proof of concept to production: a complaints investigation agent checked by LLM-as-a-judge evaluations, a GenAI customer due diligence pipeline and a reusable MCP data-access layer.
- **Before:** core engineer on a 50+ source, 1TB+/month Snowflake and dbt ESG and climate data platform whose EPC score data product is shown to customers in the NatWest app; Snowflake and PySpark analytics for Retail Decisioning; Teradata and DB2 data warehousing for Lloyds Banking Group (via TCS).
- **How I work:** hands-on individual contributor; mentoring junior engineers and setting engineering standards is part of the job.

KEY SKILLS

- Data Engineering: Python (advanced), SQL (expert), PySpark / Apache Spark (incl. AWS EMR Serverless), Apache Airflow (AWS MWAA), dbt, StreamSets, batch ELT/ETL at scale
- Data Platforms: Snowflake (architecture, warehouses, Tasks/Streams, UDFs/stored procedures, Time Travel, cloning, clustering, materialised views, performance and cost tuning), PostgreSQL, Teradata, IBM DB2, data modelling; familiar with Hadoop and Hive
- AI Data Engineering: OpenAI / Agents SDK tool calling, Kafka (consumer, familiar), MCP servers and agent tool integration, RAG / retrieval and context pipelines, LangGraph, Pydantic data contracts, LLM-as-a-judge evaluation, AI observability; AI-assisted engineering (Claude Code, Kiro CLI)
- Cloud & DevOps: AWS (S3, Glue, EMR, Athena, SageMaker), GitLab CI/CD, Docker, Git, Splunk; familiar with Terraform (maintains configuration in team repos)
- Quality & Leadership: data quality in regulated environments, data contracts, data quality gates, audit trails and lineage, automated testing, incident management; technical leadership, coaching and mentoring, onshore-offshore delivery, engineering standards, stakeholders from discovery onwards, Agile

PROFESSIONAL EXPERIENCE

Senior Data Engineer - Agentic Systems & LLM Deployment
NatWest Group, London (Hybrid) | Aug 2025 - Present

- Lead data engineer for all agentic AI products in the enterprise innovation team, from proof of concept to production, including a complaints investigation agent, a complaints decision-assist agent, an LLM-as-a-judge evaluation service and a contact-centre assist programme.
- Complaints AutoInvestigation agent: engineered the data layer for an agent that investigates a complaint across many systems and writes an investigation summary. Built on the OpenAI Agents SDK with a 30-tool registry covering core banking, credit cards, collections and case management, fed by complaints from Kafka, with tools chosen by complaint category. Every source passes data quality checks with auditing and logging, everything the agent receives is logged as facts, and encrypted outputs are stored in PostgreSQL with controls on token usage and made-up answers.
- Customer due diligence AI checks: led data engineering on a GenAI quality-check programme for commercial customer onboarding. Designed and documented a Polars-based ETL pipeline turning onboarding Kafka payloads into structured PostgreSQL datasets for LLM-driven checks, enriched with Companies House data, supporting 300+ applications a day. Built the data-quality checks, data contracts and audit trails, and wrote the handover documentation and operations runbook.
- Designed a reusable MCP server with personal-data safeguards (it returns only small slices of data, only the fields each role may see, and keeps a full audit trail) that multiple agents use with no data-access code of their own, standardising access to core banking, CRM and case-management systems.
- Built RAG / retrieval and context pipelines; traced poor retrieval results back to their source and fixed them (filtering by segment, how many candidates are considered, minimum confidence cut-offs), materially improving relevant-result recall. Cut agent data-hydration latency by keeping connections open and caching responses.
- Built PySpark pipelines on AWS EMR Serverless for model monitoring (model accuracy, LLM token usage and latency), orchestrated by Airflow.
- Introduced Snowflake CI/CD and automated testing for data loaders, and built a Snowflake writer that retries failed writes, taking deployment issues to near zero. Built Splunk observability for agent runs, tool-call performance and decision quality; deployed with GitLab CI/CD and Docker.

Senior Data Engineer - Climate Analytics (ESG Data Platform)

NatWest Group, Edinburgh | Jul 2023 - Aug 2025

- Senior engineer on the team delivering NatWest's Snowflake ESG and climate data platform on a medallion architecture: 50+ internal and external sources and 1TB+ of data a month, for analytics and regulatory reporting.
- Built the layered dbt project (RAW landing, INT integration and history, PRS presentation) with lineage, SCD-2 snapshots for vendor history, and dbt tests and docs as living documentation. Orchestrated on Airflow / AWS MWAA with config-driven pipelines, one per source, and tested builds promoted step by step to production.
- Controlled Snowflake spend by running each dbt model on a right-sized warehouse and using transient tables for rebuildable layers, cutting compute and storage cost.
- Made ingestion metadata-driven (a new source is a config row, not code) and idempotent, with pre-hook guards against duplicate loads and CloudWatch/SNS alerts that fail fast. Sped up batch processing through PySpark and dbt optimisation and reduced data quality incidents with automated monitoring and quality controls.
- Delivered climate data products (emissions, residential EPC certificates and others) used by wealth, commercial and retail teams: the EPC score is shown in the NatWest app and the data feeds climate risk scores, and portfolio views by EPC band support data-backed mortgage and green-loan propositions. Each product has a named data steward, quality metrics, data contracts between producers and consumers, and lineage back to source.
- Delivered production climate risk models with Data Architects and Data Scientists; defined engineering standards and data product frameworks; coached and mentored junior engineers on the platform and the skills they needed.

Senior Data Engineer - Retail Analytics
NatWest Group, Edinburgh | Sep 2021 - Jul 2023

- Engineered enterprise batch pipelines in Python, SQL, Snowflake, PySpark and StreamSets that move data from source to destination for the Retail Data & Analytics Decisioning team, including Customer Lifetime Value analytics, building them to be idempotent and robust.
- Designed Snowflake data models for high-volume retail analytics, using clustering, partitioning and materialised views to speed up queries for Customer Lifetime Value work. Took part in the Teradata-to-Snowflake migration, including moving data marts.

Associate Consultant - Enterprise Data Platform
Tata Consultancy Services (client: Lloyds Banking Group), Manchester & Gurugram | Jun 2019 - Sep 2021

- Moved into the enterprise data platform team to work on Teradata, and joined the data quality management unit, resolving issues flagged by overnight batch pipelines through the UI or the DB2 operational system on the mainframe.
- Worked in an onshore-offshore delivery model, mentoring team members and managing client deliveries for the Lloyds enterprise data platform: Teradata warehouse, reporting applications, other data service applications on SQL Server and Teradata, and data support for analytics teams.
- Worked in an on-premise Teradata, Hadoop, Hive and Python environment; gathered requirements for cloud migration and tried GCP and Dataform as modernisation towards GCP began.
- Supported batch runs and sought performance improvements wherever possible; took part in Teradata upgrades and disaster recovery.

Associate Consultant - Customer Information System (DB2 ODS)
Tata Consultancy Services (client: Lloyds Banking Group) | Feb 2016 - May 2019

- Supported the IBM DB2 operational data store behind the customer information system: managed data quality incidents, including customer profile merge and unmerge issues.
- Worked on data warehousing and big data across the DB2 operational data store, Teradata and Hadoop.

Software Engineer, Mainframe (Senior from 2015)
UST Global, Gurugram | Apr 2015 - Feb 2016, and **Sopra Steria**, Noida | Apr 2011 - Mar 2015

- Trained in mainframe technology, then built and supported COBOL / DB2 systems for a UK retailer and later at UST Global under ITIL processes; prepared designs with business analysts and mentored junior team members.

EDUCATION & CERTIFICATIONS

- B.Tech, Computer Science, Gautam Buddh Technical University, India (2006 - 2010)
- Certifications: Big Data on AWS (Amazon Web Services, May 2022); Data Engineering Nanodegree (Udacity, May 2022); Google Cloud Engineering Specialization (Coursera)